LEARNER GUIDE

Angles in a triangle

Angles Tell Us Why · Lesson 3. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Turn a copy of a triangle half a turn about the middle of one side, and two of its corners swap places. Do this about the middles of the two sides that meet at C, and angles a and b land at C, either side of angle c. The three angles fit together exactly on a straight line, so for every triangle **a + b + c = 180°**. An **isosceles** triangle has two equal sides and two equal base angles; an **equilateral** triangle has three equal sides and three angles of 60°; in a **right-angled** triangle the other two angles add up to 90°.

What you are learning to notice

- The three angles of every triangle fit together on a straight line: 180°.
- Two angles of a triangle always decide the third.
- Isosceles: the angles opposite the equal sides (the base angles) are equal.
- Equilateral: all three angles are 60°.
- No triangle can have two right angles, or two obtuse angles.

Moving through it, one step at a time

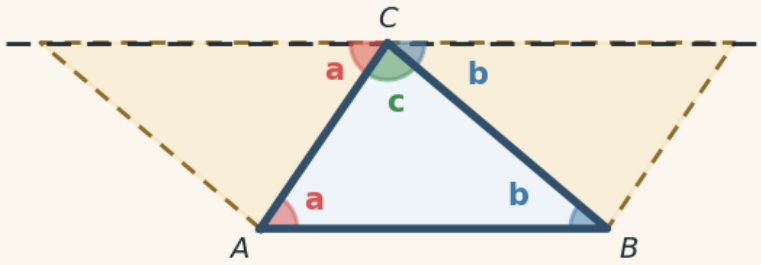
Add the angles you know, then take the total from 180°. For an isosceles triangle, look for the tick marks: the two angles at the ends of the unmarked side are equal. Write the reason with each step: “angles in a triangle add up to 180°”, “base angles of an isosceles triangle are equal”.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Why 180°?

Why do the angles of this triangle add up to 180°?

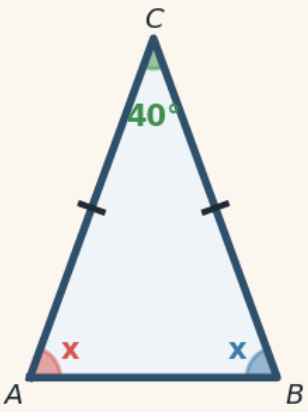


1. Turn a copy half a turn about the middle of CA.	angle a lands at C
2. Turn a copy half a turn about the middle of CB.	angle b lands at C
3. a, c and b now sit side by side on a straight line through C.	a + b + c = 180°

The turning works for every triangle, so every triangle's angles add up to 180°.

Example 2 — An isosceles triangle

The marked sides are equal. Find x.



1. The base angles are equal, so both are	x
2. Angles in a triangle add up to 180°.	$x + x + 40° = 180°$
3. So the two base angles share	$180° - 40° = 140°$
4. Each base angle is	$140° \div 2 = 70°$

x = 70°, because base angles of an isosceles triangle are equal and angles in a triangle add up to 180°.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (angles on a straight line (Lesson 2)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

> Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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